

DES: University food redistribution program using DES

Problem Definition: The university campus dining facility generates a substantial amount of surplus food each day, most of which is wasted due to inefficiencies in collection, sorting and distribution. This **food waste** represents not only an environmental issue but also a missed opportunity to contribute to food security in the surrounding community.

The main problem is optimizing the process of collecting, sorting, quality checking, and distributing surplus food to local community partners (food banks, shelters, and community kitchens). The key objectives are to:

- Minimize food waste
- Efficiently redistribute food to those in need
- Optimize resource utilization

Why DES is the most suitable approach?

- **Workflow modeling:** The food redistribution process involves multiple, sequential stages: queuing, collecting, sorting, quality checking, and distribution. Each stage is a distinct event, making Discrete Event Simulation (DES) ideal for representing the entire workflow
- **Queues and resources:** The process includes waiting times, queues, and the allocation of resources like collection staff, electric vans, and sorting stations. DES is well-suited to model these queues and resource constraints effectively, allowing us to identify bottlenecks and optimize resource allocation.
- **Performance metrics:** DES allows for the monitoring of key performance indicators (KPIs) such as waiting times at different stages, utilization of vans, and distribution time, which are essential for optimizing operations and reducing waste.

Structure and key features of DES Model

Activities

- **Surplus food collection start** (CollectionStart): Begin the food collection process from university dining halls and cafes.

- **Sorting requirement check** (SortingRequirementCheck): Decision point to determine whether the collected food requires sorting.
- **No sorting required path** (NoSortingRequired): Path for food that can skip sorting and proceed to distribution.
- **Sorting filter** (SortingFilter): Directs food items that need sorting to the sorting service.
- **Sort food items** (SortFoodItems): Sort the collected food into categories such as perishables, non-perishables, and ready-to-eat items.
- **Quality check** (QualityCheck): Check the quality of sorted food to ensure it meets safety and quality standards.
- **Not ready for distribution**: If food does not meet the quality requirements => End: disposed of, used of composting
- **Loading for distribution** (ReadyForDistribution): Load sorted and quality-checked food into electric vans for distribution.
- **Distribution start** (DistributionStart): Begin distribution to community partners such as food banks and shelters.
- **Distribution end** (DistributionEnd): Complete the distribution process and exit the system.
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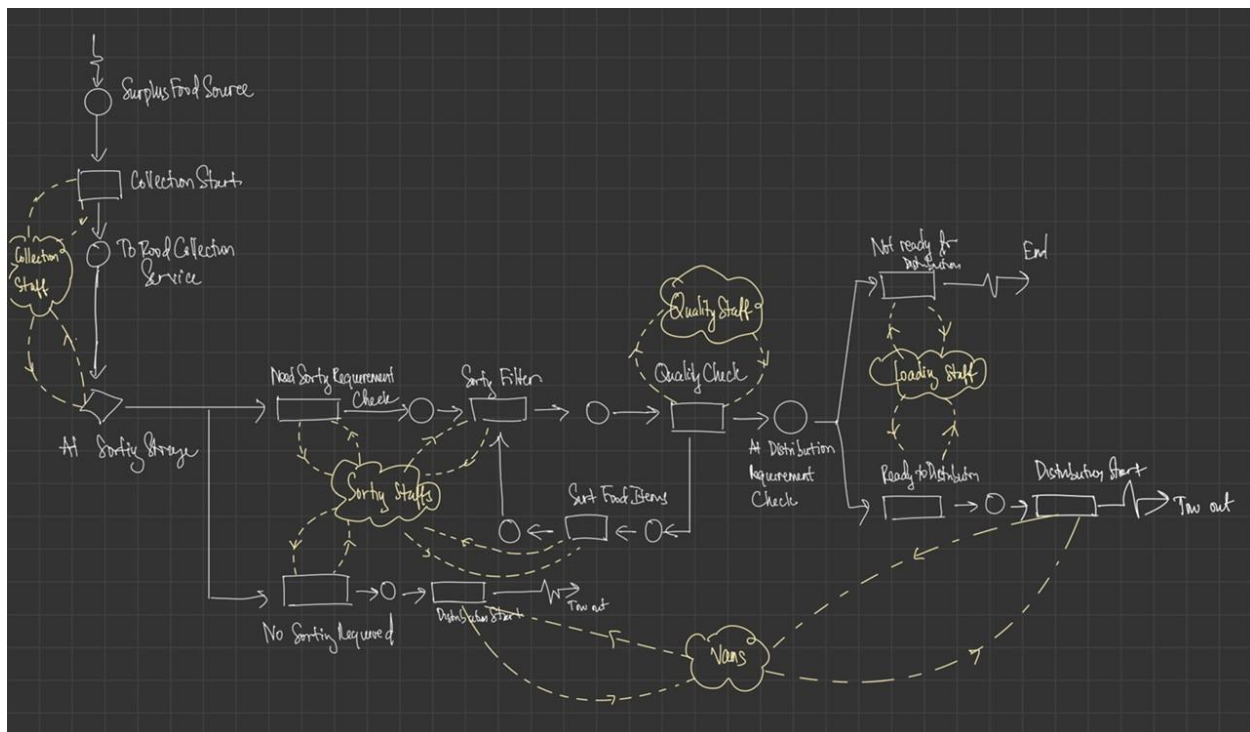
Temporary Entities

- **Surplus food**: Represents the food that is collected from various university dining facilities. These entities are temporary as they move through the system from collection to final redistribution.

Resources

- **Floating resources**:
 - **Collection staff**: These staff members are assigned to perform the initial entry check of surplus food arriving from various dining halls. => float between collection points
 - **Sorting Staff**: The sorting staff are responsible for categorizing food items and performing the quality check => float: sort food and quality check

- **Loading Staff:** Staff who assist in loading the sorted food into electric vans for redistribution => floating
- **Dedicated resources:**
 - **Electric vans:** Vans used for transporting collected food either from the collection points to the sorting center or for final delivery to community partners. Each van is assigned to specific tasks and food types.
 - **Quality staff:** Perform quality checks on sorted food



Overview of the DES Flow

- **Surplus Food Collection:** Food is **generated** from university dining halls and cafes and enters the collection process.
 - **Collection staff** float between various collection points to bring the surplus food to the central processing area.
- **Sorting Requirement Check:** A **decision point** determines whether the food needs sorting. If it does not, it proceeds directly to **Loading for Distribution**.

⇒ If **sorting is required**, the food is directed to **Sort Food Items**.

- **Sorting and Quality Checks:**

- **Sorting Staff** categorize food based on perishability and type (perishable, non-perishable, ready-to-eat).
- **Quality Staff** then perform a thorough **quality check** to confirm suitability for redistribution.
- Items failing the quality check are routed to **Not Ready for Distribution** for disposal or composting.

- **Loading and Distribution:**

- **Approved food** is loaded into electric vans by the **Loading Staff**.
- Vans then start the distribution journey to community partners such as **food banks** and **shelters**.

- **Distribution End:** Once delivered, food exits the system, completing the redistribution cycle.

How to Acquire Data for the Parameters of the DES Model

1. Surplus Food Generation Rate (SurplusFoodSource)

- **Data Source:**

- Gather data from **university dining halls and cafes** on the daily volume of surplus food generated. This can be done by maintaining a **daily logbook** that records the amount of leftover food at the end of each day.
- Use **kitchen management software** or **manual logs** to track the amount of food that is not served.

- **Data Collection Method:**

- **Direct Measurements:** Weigh surplus food at each dining facility daily.

- **Historical Data Analysis:** Look at records over several months to calculate average surplus generation per day and analyze seasonal variations (e.g., during exams, vacations).

2. Collection Times (CollectionStart and ToRoadCollectionService)

- **Data Source:**

- The time required to **collect surplus food** from each collection point (dining hall or cafe) to the central facility.

- **Data Collection Method:**

- **Observation:** Conduct **time studies** by directly measuring the time taken for each trip by collection staff.
- **GPS Tracking:** Use GPS to monitor the routes and time taken by collection staff when moving food between collection points and the central sorting area.
- **Interviews with Collection Staff:** Gather insights from the staff on factors that might cause variations in collection times, such as distance between halls and time of day.

3. Sorting Requirement Check and Sorting Time (SortingRequirementCheck, SortFoodItems)

- **Data Source:**

- **Sorting Requirement Check** requires data to understand the proportion of food needing sorting.
- **Sorting Time** involves data on how long it takes to sort different types of food.

- **Data Collection Method:**

- **Categorization Observations:** Use a sample of surplus food collected over a week to determine the **proportion of food items** that need sorting versus those that do not. This gives you an average proportion to be used in the decision point (SortingRequirementCheck).

- **Time-Motion Studies:** Observe **sorting staff** while they categorize the surplus food. This will provide an average sorting time for each category—**perishables, non-perishables, ready-to-eat meals**.
- **Historical Records:** Collect historical records of the average number of items processed by each sorting staff per hour.

4. Quality Check Times (QualityCheck)

- **Data Source:**
 - The **quality check process** includes evaluating sorted food to determine if it is suitable for distribution.
- **Data Collection Method:**
 - **Observation:** Measure the time taken for the **Quality Staff** to check the quality of food items. This can be done by timing multiple quality checks on different food categories (e.g., perishables vs. non-perishables).
 - **Volunteer Input:** Interview quality check volunteers to understand **variability in time**, such as additional checks required for borderline quality items.

5. Not Ready for Distribution (Disposal/Composting)

- **Data Source:**
 - The **rate at which food is deemed unsuitable** for redistribution and must be disposed of or used for composting.
- **Data Collection Method:**
 - **Quality Records:** Maintain records on food items that **fail the quality check**. Track these over several weeks to estimate the proportion of food that is unsuitable for distribution.
 - **Feedback from Sorting and Quality Staff:** Use input from sorting and quality check staff to identify patterns in food rejection (e.g., based on specific food types or times of year).

6. Loading Time for Distribution (ReadyForDistribution)

- **Data Source:**

- The time taken to load **approved food** into the electric vans, considering various types of food (e.g., perishables and ready-to-eat).

- **Data Collection Method:**

- **Observation:** Conduct **loading time studies** to determine the time needed by loading staff to move sorted food into vans.
- **Staff Feedback:** Collect insights from the loading staff about **challenges** they face during loading (e.g., handling heavier or bulkier items).
- **Equipment Utilization:** Measure how the availability of loading tools (such as trolleys or lifts) impacts loading times.

7. Distribution Time (DistributionStart and DistributionEnd)

- **Data Source:**

- The time taken for **electric vans** to distribute food to community partners, such as food banks and shelters.

- **Data Collection Method:**

- **GPS Tracking:** Track **electric vans** to monitor the routes, stops, and time taken to complete deliveries. Use the data to determine **average travel time** based on traffic, distance, and other influencing factors.
- **Historical Data:** Use historical delivery records to understand typical delivery times for different partner locations.
- **Delivery Route Optimization Software:** Use routing software to identify potential **optimizations** and measure the differences between the suggested route times and actual delivery times.

8. Queue Lengths and Waiting Times (All Stages)

- **Data Source:**

- Waiting times at **collection, sorting, quality check, and loading** stages to identify bottlenecks and improve efficiency.

- **Data Collection Method:**

- **Manual Counting:** Count the **number of food items** waiting at each stage during peak and non-peak hours to identify typical queue lengths.
- **RFID Tags:** Use RFID tagging on food containers to track how long they spend in each **waiting queue**.
- **Digital Time Logging:** Use a **time log** software tool to record when food items enter and leave each stage.

Summary of Data Collection for DES Model Parameters

Activity	Parameter	Data Collection Method
Collection Start	Collection Times	Observation, GPS Tracking, Staff Interviews
Sorting Requirement Check	Proportion of Food Needing Sorting	Categorization Observations
Sort Food Items	Sorting Time	Time-Motion Studies, Historical Records
Quality Check	Quality Check Time	Observation, Volunteer Input
Not Ready for Distribution	Proportion of Food Rejected	Quality Records, Staff Feedback
Loading for Distribution	Loading Time	Observation, Staff Feedback
Distribution Start/End	Distribution Time	GPS Tracking, Historical Data
Queues and Waiting Times	Queue Lengths and Waiting Times	Manual Counting, RFID Tags, Digital Logging